

CSA5103 - CRYPTOGRAPHY AND NETWORK SECURITY LIST OF PROGRAMS

EXPERIMENT 2

Write a program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

Aim: To write a Python program that encrypts and decrypts a message using the Monoalphabetic Substitution Cipher.

Algorithm:

Step 1: Define the plaintext alphabet and the corresponding ciphertext alphabet.

Step 2: Read the plaintext message from the user.

Step 3: Convert each plaintext character to its corresponding ciphertext character using the substitution alphabet.

Step 4: Display the encrypted (ciphertext) message.

Step 5: Convert each ciphertext character back to its corresponding plaintext character.

Step 6: Display the decrypted (original) message.

Step 7: End.

Program:

```
plain = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
cipher = "QWERTYUIOPASDFGHJKLZXCVBNM"
plaintext = input("Enter the plaintext: ").upper()
ciphertext = ""
for ch in plaintext:
    if ch in plain:
        index = plain.index(ch)
        ciphertext += cipher[index]
    else:
        ciphertext += ch
print("Encrypted Text:", ciphertext)
decrypted = ""
for ch in ciphertext:
    if ch in cipher:
        index = cipher.index(ch)
        decrypted += plain[index]
    else:
        decrypted += ch
print("Decrypted Text:", decrypted)
```

Output:

```
Enter the plaintext: HELLO WORLD
Encrypted Text: ITSSG VGKSR
Decrypted Text: HELLO WORLD
```

Result: The Monoalphabetic Substitution Cipher was implemented successfully in Python